

TRAVAUX DIRIG  S N   2 : Empirical Risk, Concentration, VC Theory

Stephan CL  MEN  ON <stephan.clemencon@telecom-paris.fr>
Quentin BOUNIOT <quentin.bouniot@telecom-paris.fr>

EXERCICE 1. We work in the framework of binary classification : let X be a random descriptor taking values in a measurable space $\mathcal{X} \subset \mathbb{R}^d$ ($d \in \mathbb{N}^*$), and let Y be a random label taking the value -1 or 1 . Consider a finite class $\mathcal{G}$ of classifiers $\mathcal{X} \rightarrow \{-1, 1\}$ such that the two labels are perfectly separable by an element of $\mathcal{G}$, i.e. $\min_{g \in \mathcal{G}} L(g) = 0$, for the risk $L : g \in \mathcal{G} \mapsto \mathbb{P}(g(X) \neq Y) \in [0, 1]$.

Let $n \in \mathbb{N}^*$. Suppose that we have an i.i.d. sample $\{(X_i, Y_i)\}_{1 \leq i \leq n}$ following the same distribution as (X, Y) , and let $\hat{g}_n$ be a minimizer of the empirical classification error :

$$\hat{g}_n \in \min_{g \in \mathcal{G}} L_n(g) \quad \text{where} \quad L_n : g \in \mathcal{G} \mapsto \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{g(X_i) \neq Y_i\}}.$$

- 1) Show that $\min_{g \in \mathcal{G}} L_n(g) = 0$ almost surely.
- 2) Show that $\mathbb{P}(L(\hat{g}_n) > \epsilon) \leq |\mathcal{G}|(1 - \epsilon)^n$ for every $\epsilon \in [0, 1]$.
Deduce that $\mathbb{P}(L(\hat{g}_n) > \epsilon) \leq |\mathcal{G}|e^{-n\epsilon}$.

Hint. Use $\mathcal{G}_\epsilon := \{g \in \mathcal{G} : L(g) > \epsilon\}$ together with a union bound.

- 3) Deduce from the previous question that $\mathbb{E}(L(\hat{g}_n)) \leq \frac{\log(e|\mathcal{G}|)}{n}$.

Hint. For every positive random variable Z , $\mathbb{E}(Z) = \int_0^{+\infty} \mathbb{P}(Z > t) dt$.

Correction Exercise 1

Here we assume $\min_{g \in \mathcal{G}} L(g) = 0$, which means that $g^* \in \mathcal{G}$ and $L(g^*) = 0$. This is a strong assumption, but it allows us to derive some theoretical results. Also note that $\hat{g}_n$ is a random variable since it depends on the random samples $\{(X_i, Y_i)\}_{1 \leq i \leq n}$.

- 1) From the hypothesis, $\exists g^* \in \mathcal{G}$ such that $L(g^*) = 0$.

$$\begin{aligned} \mathbb{P}(L_n(g^*) = 0) &= \mathbb{P}\left(\sum_{i=1}^n \mathbb{1}_{\{Y_i \neq g^*(X_i)\}} = 0\right) \\ &= \mathbb{P}(Y_1 = g^*(X_1), \dots, Y_n = g^*(X_n)) \\ &= \mathbb{P}(Y = g^*(X))^n \quad ((X_i, Y_i) \sim (X, Y) \text{ and i.i.d.}) \\ &= (1 - L(g^*))^n = 1 \end{aligned}$$

Since $0 \leq \min_{g \in \mathcal{G}} L_n(g) \leq L_n(g^*)$, we get $\min_{g \in \mathcal{G}} L_n(g) = 0$ almost surely.

- 2) Let $n \in \mathbb{N}$ and $\epsilon \in \mathbb{R}_+$. Note that $\forall g \in \mathcal{G}, L(g) \in [0, 1]$, thus for $\epsilon \geq 1$, $\mathbb{P}(L(\hat{g}_n) > \epsilon) = 0$. So

we can assume $\epsilon \in [0, 1[$. Let $\mathcal{G}_\epsilon := \{g \in \mathcal{G} : L(g) > \epsilon\}$. Then we have :

$$\begin{aligned}
\mathbb{P}(L(\hat{g}_n) > \epsilon) &= \mathbb{P}(\hat{g}_n \in \mathcal{G}_\epsilon) \\
&= \mathbb{P}(\hat{g}_n \in \mathcal{G}_\epsilon, L_n(\hat{g}_n) = 0) \quad (\text{by definition of } \hat{g}_n \text{ and q.1}) \\
&\leq \mathbb{P}\left(\bigcup_{g \in \mathcal{G}_\epsilon} \{L_n(g) = 0\}\right) \quad (\text{event for } \hat{g}_n \text{ is included in the union}) \\
&\leq \sum_{g \in \mathcal{G}_\epsilon} \mathbb{P}(L_n(g) = 0) \quad (\text{union bound, } \mathcal{G} \text{ is finite}) \\
&= \sum_{g \in \mathcal{G}_\epsilon} \mathbb{P}(g(X) = Y)^n \quad (\text{same as q.1 : samples are iid from } (X, Y)) \\
&= \sum_{g \in \mathcal{G}_\epsilon} (1 - L(g))^n \\
&\leq \sum_{g \in \mathcal{G}_\epsilon} (1 - \epsilon)^n \quad (\text{since } L(g) > \epsilon) \\
&= |\mathcal{G}_\epsilon| (1 - \epsilon)^n
\end{aligned}$$

Finally, by convexity of the exponential function, we have $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, thus $(1 - \epsilon)^n \leq e^{-n\epsilon}$, and $\mathbb{P}(L(\hat{g}_n) > \epsilon) \leq |\mathcal{G}_\epsilon| e^{-n\epsilon} \leq |\mathcal{G}| e^{-n\epsilon}$ because $|\mathcal{G}_\epsilon| \leq |\mathcal{G}|$.

3) Let $Z = L(\hat{g}_n)$, which is a positive random variable (because of the random samples). Then, $\mathbb{P}(Z > \epsilon) \leq \min(1, |\mathcal{G}| e^{-n\epsilon})$ (since $P(Z) \in [0, 1]$), and $|\mathcal{G}| e^{-n\epsilon} \leq 1$ for $\epsilon \geq \frac{\ln(|\mathcal{G}|)}{n}$. Thus, we can write :

$$\begin{aligned}
\mathbb{E}(Z) &= \int_0^{+\infty} \mathbb{P}(Z > \epsilon) d\epsilon \\
&\leq \int_0^{\frac{\ln(|\mathcal{G}|)}{n}} 1 d\epsilon + \int_{\frac{\ln(|\mathcal{G}|)}{n}}^{+\infty} |\mathcal{G}| e^{-n\epsilon} d\epsilon \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + \int_{\frac{\ln(|\mathcal{G}|)}{n}}^{+\infty} |\mathcal{G}| e^{-n\epsilon} d\epsilon \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + |\mathcal{G}| \int_{\frac{\ln(|\mathcal{G}|)}{n}}^{+\infty} e^{-n\epsilon} d\epsilon \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + |\mathcal{G}| \left[-\frac{1}{n} e^{-n\epsilon} \right]_{\frac{\ln(|\mathcal{G}|)}{n}}^{+\infty} \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + |\mathcal{G}| \left(0 + \frac{1}{n} e^{-n \cdot \frac{\ln(|\mathcal{G}|)}{n}} \right) \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + |\mathcal{G}| \left(\frac{1}{n} \cdot \frac{1}{|\mathcal{G}|} \right) \\
&\leq \frac{\ln(|\mathcal{G}|)}{n} + \frac{1}{n} \\
&\leq \frac{\ln(|\mathcal{G}|) + 1}{n} = \frac{\ln(e|\mathcal{G}|)}{n}
\end{aligned}$$

We could also obtain a tighter bound by integrating from 0 to 1 directly since $Z \in [0, 1]$:

$$\mathbb{E}(Z) \leq \frac{\ln(|\mathcal{G}|)}{n} + \int_{\frac{\ln(|\mathcal{G}|)}{n}}^1 |\mathcal{G}| e^{-n\epsilon} d\epsilon = \frac{\ln(|\mathcal{G}|)}{n} + \frac{1}{n} - \frac{|\mathcal{G}|}{n} e^{-n}.$$

EXERCICE 2. We work in the framework of binary classification. We use the same notation as in the

previous exercise. Set $L^* := L(g^*)$, where $g^* : x \in \mathcal{X} \mapsto 2\mathbb{1}_{\{\eta(x) \geq 1/2\}} - 1$, and let $\eta : x \in \mathcal{X} \mapsto \mathbb{P}(Y = 1 | X = x) \in [0, 1]$ be the regression function. Let $(\eta_n)_{n \in \mathbb{N}^*}$ be a sequence of functions defined on $\mathcal{X}$ with values in $]0, 1[$. For every $n \in \mathbb{N}^*$, consider the classifier $g_n : x \in \mathcal{X} \mapsto 2\mathbb{1}_{\{\eta_n(x) \geq 1/2\}} - 1$.

- 1) Suppose that there exists $\delta > 0$ such that $|\eta(x) - 1/2| \geq \delta$ for every $x \in \mathcal{X}$. Show that

$$L(g_n) - L^* \leq \frac{2 \mathbb{E} ((\eta_n(X) - \eta(X))^2)}{\delta}.$$

- 2) Show that if $L^* = 0$, then for every $q \in [1, +\infty[$,

$$L(g_n) \leq 2^q \mathbb{E} (|\eta_n(X) - \eta(X)|^q).$$

Now let $\eta' : \mathcal{X} \rightarrow]0, 1[$ and $g' : x \in \mathcal{X} \mapsto 2\mathbb{1}_{\{\eta'(x) \geq 1/2\}} - 1$.

- 3) Suppose that $\mathbb{P}\{\eta'(X) = 1/2\} = 0$ and that $\mathbb{E} (|\eta_n(X) - \eta'(X)|) \rightarrow 0$ as $n \rightarrow +\infty$. Show that $L(g_n) \rightarrow L(g')$ as $n \rightarrow +\infty$.
- 4) Suppose that the label Y is no longer observable, but that a variable Z taking values in $\{-1, +1\}$ is observable, such that :

$$\begin{aligned} \mathbb{P}(Z = 1 | Y = -1, X) &= \mathbb{P}(Z = 1 | Y = -1) = a < 1/2, \\ \mathbb{P}(Z = -1 | Y = 1, X) &= \mathbb{P}(Z = -1 | Y = 1) = b < 1/2. \end{aligned}$$

Now set $\eta' : x \in \mathcal{X} \mapsto \mathbb{P}(Z = +1 | X = x)$. Show that :

$$L(g') \leq L^* \left(1 + \frac{2|a - b|}{1 - 2\max(a, b)} \right).$$

What can be deduced when $a = b$?

Correction Exercise 2

1) First, we can notice that the inequality is trivial if $L(g_n) = L^*$, so we can assume $L(g_n) \neq L^*$. Then, we have :

$$\begin{aligned} L(g_n) - L^* &= \mathbb{E}[\mathbb{1}_{g_n(X) \neq Y} - \mathbb{1}_{g^*(X) \neq Y}] \\ &= \mathbb{E}[\eta(X) [\mathbb{1}_{g_n(X) = -1} - \mathbb{1}_{g^*(X) = -1}] + (1 - \eta(X)) [\mathbb{1}_{g_n(X) = 1} - \mathbb{1}_{g^*(X) = 1}]] \quad (\text{total exp.}) \\ &= \mathbb{E}[(2\eta(X) - 1) [\mathbb{1}_{g_n(X) = -1} - \mathbb{1}_{g^*(X) = -1}]] \quad (\mathbb{1}_A = 1 - \mathbb{1}_{\bar{A}}) \end{aligned}$$

Let $\forall x \in \mathcal{X}, h(x) := \mathbb{1}_{g_n(X) = -1} - \mathbb{1}_{g^*(X) = -1} \in \{-1, 0, 1\}$, we can then write :

$$L(g_n) - L^* = 2\mathbb{E}[(\eta(X) - 1/2)h(X)]. \quad (1)$$

Let's look at the cases :

- $h(x) \leq 0$ if $\eta(x) < 1/2$,
- $h(x) \geq 0$ if $\eta(x) > 1/2$.

Thus, $\forall x \in \mathcal{X}, (\eta(x) - 1/2) \cdot h(x) = |\eta(x) - 1/2| \cdot |h(x)| \geq 0$, and, $|h(x)| = \mathbb{1}_{g_n(x) \neq g^*(x)}$.

Then, $\forall x \in \mathcal{X}$, if $g_n(x) \neq g^*(x)$, then :

- if $\eta(x) \geq 1/2$, then $\eta_n(x) < 1/2 \leq \eta(x)$, and $1/2 - \eta(x) > \eta_n(x) - \eta(x)$,
- if $\eta(x) < 1/2$, then $\eta_n(x) \geq 1/2 > \eta(x)$, and $\eta(x) - 1/2 > \eta(x) - \eta_n(x)$.

Thus, $|\eta_n(x) - \eta(x)| \geq |\eta(x) - 1/2|$. Finally, if $g_n(x) = g^*(x)$, then $h(x) = 0$, and we can summarize the result as :

$$\forall x \in \mathcal{X}, |h(x)| \cdot |\eta(x) - 1/2| \leq |h(x)| \cdot |\eta_n(x) - \eta(x)|. \quad (2)$$

By assumption, we have $|\eta(x) - 1/2| \geq \delta$, for $\delta > 0$, thus inserting this in (1) :

$$L(g_n) - L^* \geq 2\delta \mathbb{E}[|h(X)|]. \quad (3)$$

Now inserting (2) in (1), we have :

$$\begin{aligned} L(g_n) - L^* &\leq 2\mathbb{E}[|h(X)| \cdot |\eta_n(X) - \eta(X)|] \\ &\leq 2\sqrt{\mathbb{E}[|h(X)|^2] \cdot \mathbb{E}[|\eta_n(X) - \eta(X)|^2]} \quad (\text{Cauchy-Schwarz}) \\ &\leq 2\sqrt{\mathbb{E}[|h(X)|] \cdot \mathbb{E}[|\eta_n(X) - \eta(X)|^2]} \quad (|h(X)| \in [0, 1]) \\ &\leq 2\sqrt{\frac{L(g_n) - L^*}{2\delta} \cdot \mathbb{E}[|\eta_n(X) - \eta(X)|^2]} \quad (\text{from (3)}) \end{aligned}$$

Finally, by squaring both sides, and since $L(g_n) - L^* \neq 0$, we have :

$$L(g_n) - L^* \leq \frac{2}{\delta} \mathbb{E}[|\eta_n(X) - \eta(X)|^2].$$

2) If $L^* = 0$, then $g^*(X) = Y$ almost-surely (by definition of L^*). If $L(g_n) = 0$, then the inequality is trivial, so we can assume $L(g_n) \neq 0$.

Let's start back from (1) after inserting (2) :

$$\begin{aligned} L(g_n) &\leq 2\mathbb{E}[|\eta_n(X) - \eta(X)| \cdot |h(X)|] \\ &\leq 2\mathbb{E}[|\eta_n(X) - \eta(X)|^q]^{1/q} \cdot \mathbb{E}[|h(X)|^{q/(q-1)}]^{(q-1)/q} \quad (\text{Hölder's inequality, } \forall q > 1) \end{aligned}$$

Then, since $|h(X)| \in [0, 1]$, we have $|h(X)|^{q/(q-1)} = |h(X)|$, and, $|h(X)| = \mathbb{1}_{g_n(X) \neq g^*(X)} = \mathbb{1}_{g_n(X) \neq Y}$ (almost-surely), and thus $\mathbb{E}(|h(X)|) = L(g_n)$. Finally, we have :

$$\begin{aligned} L(g_n) &\leq 2\mathbb{E}[|\eta_n(X) - \eta(X)|^q]^{1/q} \cdot L(g_n)^{(q-1)/q} \\ L(g_n) &\leq 2^q \mathbb{E}[|\eta_n(X) - \eta(X)|^q] \quad (L(g_n) \neq 0) \end{aligned}$$

The case $q = 1$ follows directly from the first line, since $|h(X)| \leq 1$.

3) From the same computation as in question 1 :

$$\begin{aligned} |L(g_n) - L(g')| &= 2|\mathbb{E}[(\eta(X) - 1/2)(\mathbb{1}_{g_n(X)=-1} - \mathbb{1}_{g'(X)=-1})]| \\ &\leq 2\mathbb{E}[|\eta(X) - 1/2| \cdot |\mathbb{1}_{g_n(X)=-1} - \mathbb{1}_{g'(X)=-1}|] \quad (\text{triangular inequality}) \\ &\leq \mathbb{E}[|\mathbb{1}_{g_n(X)=-1} - \mathbb{1}_{g'(X)=-1}|] \quad (|\eta(X) - 1/2| \leq 1/2) \\ &\leq \mathbb{P}(g_n(X) \neq g'(X)) \\ &= \mathbb{P}(\eta_n(X) \geq 1/2, \eta'(X) < 1/2) + \mathbb{P}(\eta_n(X) < 1/2, \eta'(X) \geq 1/2) \end{aligned}$$

Let $\lambda \in]0, 1/2[$, we can split the events regarding $\eta'(X)$ into :

- $\{\eta'(X) < 1/2\} = \{\eta'(X) \leq 1/2 - \lambda\} \cup \{\eta'(X) \in [1/2 - \lambda, 1/2[\}$,
- $\{\eta'(X) \geq 1/2\} = \{\eta'(X) \geq 1/2 + \lambda\} \cup \{\eta'(X) \in [1/2, 1/2 + \lambda[\}$.

Then if we decompose the cartesian product of the events, we have :

- $\{\eta_n(X) \geq 1/2\} \cap \{\eta'(X) \leq 1/2 - \lambda\} \subset \{|\eta_n(X) - \eta'(X)| \geq \lambda\}$,
- $\{\eta_n(X) \geq 1/2\} \cap \{\eta'(X) \in [1/2 - \lambda, 1/2[\} \subset \{|\eta'(X) - 1/2| \leq \lambda\}$,
- $\{\eta_n(X) \leq 1/2\} \cap \{\eta'(X) \geq 1/2 + \lambda\} \subset \{|\eta_n(X) - \eta'(X)| \geq \lambda\}$,
- $\{\eta_n(X) \leq 1/2\} \cap \{\eta'(X) \in [1/2, 1/2 + \lambda[\} \subset \{|\eta'(X) - 1/2| \leq \lambda\}$.

This gives us :

$$|L(g_n) - L(g')| \leq 2\mathbb{P}(|\eta_n(X) - \eta'(X)| \geq \lambda) + 2\mathbb{P}(|\eta'(X) - 1/2| \leq \lambda) \quad (4)$$

Let $\epsilon > 0$. We start with the second term :

Since $\mathbb{P}(\eta'(X) = 1/2) = 0$, we have $\lim_{\lambda \rightarrow 0} \mathbb{P}(|\eta'(X) - 1/2| \leq \lambda) = 0$ since $\{|\eta'(X) - 1/2| \leq \lambda\}$ decreases (from the point of view of included sets) to $\{\eta'(X) = 1/2\}$ as $\lambda \rightarrow 0$, and $\mathbb{P}$ is a continuous measure. Thus, there exists $\lambda_0 > 0$ such that $\mathbb{P}(|\eta'(X) - 1/2| \leq \lambda_0) < \epsilon/4$.

For the first term, we have :

$$\mathbb{P}(|\eta_n(X) - \eta'(X)| \geq \lambda_0) \leq \frac{\mathbb{E}(|\eta_n(X) - \eta'(X)|)}{\lambda_0} \quad (\text{Markov's inequality}).$$

Since $\mathbb{E}(|\eta_n(X) - \eta'(X)|) \rightarrow 0$ as $n \rightarrow +\infty$, there exists $N_0 \in \mathbb{N}$ such that for all $n \geq N_0$, $\mathbb{E}(|\eta_n(X) - \eta'(X)|) < \lambda_0 \epsilon/4$, and thus $\mathbb{P}(|\eta_n(X) - \eta'(X)| \geq \lambda_0) < \epsilon/4$.

Plugging these two results into (4), we showed that $L(g_n) \rightarrow L(g')$ as $n \rightarrow +\infty$.

4) First, for every $x \in \mathcal{X}$, by the law of total probability and the assumptions on Z :

$$\begin{aligned} \eta'(x) &= \mathbb{P}(Z = 1 \mid X = x) \\ &= \mathbb{P}(Z = 1 \mid Y = 1, X = x) \mathbb{P}(Y = 1 \mid X = x) + \mathbb{P}(Z = 1 \mid Y = -1, X = x) \mathbb{P}(Y = -1 \mid X = x) \\ &= (1 - b) \eta(x) + a (1 - \eta(x)) \\ &= a + (1 - a - b) \eta(x). \end{aligned}$$

Note that $1 - a - b > 0$ since $a, b < 1/2$, so η' is an increasing affine transformation of η .

The computations of question 1 leading to (1) and (2) hold for any classifier of the form $x \mapsto 2\mathbb{1}_{\{\tilde{\eta}(x) \geq 1/2\}} - 1$, so we can apply them with η' instead of η_n :

$$L(g') - L^* \leq 2 \mathbb{E} \left[|\eta'(X) - \eta(X)| \mathbb{1}_{\{g'(X) \neq g^*(X)\}} \right].$$

We need to bound $|\eta'(X) - \eta(X)|$ on the event $\{g'(X) \neq g^*(X)\}$, and then the probability of this event.

Bound on $|\eta' - \eta|$. Let $t := \frac{1-2a}{2(1-a-b)}$. Since $\eta'(x) = a + (1 - a - b)\eta(x)$, we have $\eta'(x) \geq 1/2 \iff \eta(x) \geq t$. Moreover, $t - \frac{1}{2} = \frac{b-a}{2(1-a-b)}$. Let $x \in \mathcal{X}$ such that $g'(x) \neq g^*(x)$. There are two cases :

(i) $g^*(x) = 1$ and $g'(x) = -1$, i.e. $\frac{1}{2} \leq \eta(x) < t$, which is only possible if $a < b$. Then

$$|\eta'(x) - \eta(x)| = \eta(x) - \eta'(x) = (a+b) \eta(x) - a < (a+b) t - a = \frac{b-a}{2(1-a-b)} = \frac{|a-b|}{2(1-a-b)}.$$

(ii) $g^*(x) = -1$ and $g'(x) = 1$, i.e. $t \leq \eta(x) < \frac{1}{2}$, which is only possible if $a > b$. Then

$$|\eta'(x) - \eta(x)| = \eta'(x) - \eta(x) = a - (a+b) \eta(x) \leq a - (a+b) t = \frac{a-b}{2(1-a-b)} = \frac{|a-b|}{2(1-a-b)}.$$

Thus, in both cases $|\eta'(x) - \eta(x)| \leq \frac{|a-b|}{2(1-a-b)}$, and :

$$L(g') - L^* \leq \frac{|a-b|}{1-a-b} \mathbb{P}(g'(X) \neq g^*(X)). \quad (5)$$

Bound on $\mathbb{P}(g'(X) \neq g^*(X))$. Recall (TD1) that $L^* = \mathbb{E} [\min(\eta(X), 1 - \eta(X))]$. On the event $\{g'(X) \neq g^*(X)\}$:

- in case (i) ($a < b$) : $\eta(X) \in [1/2, t[$, so $\min(\eta(X), 1 - \eta(X)) = 1 - \eta(X) > 1 - t = \frac{1-2b}{2(1-a-b)}$,
- in case (ii) ($a > b$) : $\eta(X) \in [t, 1/2[$, so $\min(\eta(X), 1 - \eta(X)) = \eta(X) \geq t = \frac{1-2a}{2(1-a-b)}$.

In both cases, $\min(\eta(X), 1 - \eta(X)) \geq \frac{1-2\max(a,b)}{2(1-a-b)}$ on $\{g'(X) \neq g^*(X)\}$ (and this event is empty if $a = b$). Hence :

$$L^* \geq \mathbb{E} [\min(\eta(X), 1 - \eta(X)) \mathbb{1}_{\{g'(X) \neq g^*(X)\}}] \geq \frac{1-2\max(a,b)}{2(1-a-b)} \mathbb{P}(g'(X) \neq g^*(X)),$$

and, since $1 - 2\max(a, b) > 0$:

$$\mathbb{P}(g'(X) \neq g^*(X)) \leq \frac{2(1-a-b)}{1-2\max(a,b)} L^*.$$

Conclusion. Plugging this into (5) :

$$L(g') - L^* \leq \frac{|a-b|}{1-a-b} \cdot \frac{2(1-a-b)}{1-2\max(a,b)} L^* = \frac{2|a-b|}{1-2\max(a,b)} L^*,$$

which gives the result : $L(g') \leq L^* \left(1 + \frac{2|a-b|}{1-2\max(a,b)}\right)$.

When $a = b$, we have $t = 1/2$, so $\eta'(x) \geq 1/2 \iff \eta(x) \geq 1/2$, i.e. $g' = g^*$ and $L(g') = L^*$: the Bayes classifier of the noisy problem is the Bayes classifier of the original problem. Symmetric label noise does not change the optimal decision rule.

EXERCISE 3. Compute the VC dimension of the following classes $\mathcal{A}$ of sets :

- 1) $\mathcal{A} = \{] - \infty, x_1] \times \dots \times] - \infty, x_d] : (x_1, \dots, x_d) \in \mathbb{R}^d \}$,
- 2) $\mathcal{A}$ is the class of rectangles in $\mathbb{R}^d$.

Correction Exercise 3

1) Let's start with some examples to build an intuition :

For $d = 1$: we can shatter one point z_1 , but we cannot shatter two points z_1, z_2 . Thus, $N_{\mathcal{A}}(z_1) = 2$, $N_{\mathcal{A}}(z_1, z_2) = 3 < 2^2$, $S_{\mathcal{A}}(2) = 3 < 2^2$, and the VC dimension is $V_{\mathcal{A}} = 1$.

For $d = 2$: we can shatter two points z_1, z_2 , but we cannot shatter three points z_1, z_2, z_3 . Thus, $N_{\mathcal{A}}(z_1) = 2$, $N_{\mathcal{A}}(z_1, z_2) = 4$, $N_{\mathcal{A}}(z_1, z_2, z_3) = 7 < 2^3$, $S_{\mathcal{A}}(3) = 7 < 2^3$, and the VC dimension is $V_{\mathcal{A}} = 2$.

Now we want to prove that for any $d \geq 1$, the VC dimension is $V_{\mathcal{A}} = d$. For any $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, we can write $A_x =] - \infty, x_1] \times \dots \times] - \infty, x_d]$ such that $\mathcal{A} = \{A_x : x \in \mathbb{R}^d\}$. Let $z_1, \dots, z_n \in \mathbb{R}^d$ be n points. We write $z_i = (z_{i,1}, \dots, z_{i,d})$. We need to show :

- There exists $z_1, \dots, z_d$ such that $N_{\mathcal{A}}(z_1, \dots, z_d) = 2^d$ (i.e., we can shatter d points),

— For any $z_1, \dots, z_{d+1}$, we have $N_{\mathcal{A}}(z_1, \dots, z_{d+1}) < 2^{d+1}$ (i.e., we cannot shatter $d + 1$ points).

Let's first find a set of d points that can be shattered. We can take the points $e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0), \dots, e_d = (0, \dots, 0, 1)$ (the standard / canonical basis vectors of $\mathbb{R}^d$).

$$\forall i \in \{1, \dots, d\}, x \in \mathbb{R}^d, \quad e_i \in A_x \iff x_i \geq 1 \text{ and } x_j \geq 0 \text{ for all } j \neq i.$$

Then, for any $x \in \mathbb{R}^d$ (e.g. $x = 0$),

$$\{e_1, \dots, e_d\} \cap A_x = \emptyset \iff \text{for all } i, x_i < 1 \text{ or there exists } j \neq i \text{ such that } x_j < 0.$$

Finally, for any $m \in \{1, \dots, d\}$ and $i_1, \dots, i_m \in \{1, \dots, d\}$ distinct, we have :

$$x_{i_1}, \dots, x_{i_m} \geq 1 \text{ and } 0 \leq x_j < 1 \text{ for all } j \notin \{i_1, \dots, i_m\} \implies \{e_1, \dots, e_d\} \cap A_x = \{e_{i_1}, \dots, e_{i_m}\}.$$

With this, we can see that we can obtain all the 2^d possible intersections. Thus, $N_{\mathcal{A}}(e_1, \dots, e_d) = 2^d$, $S_{\mathcal{A}}(d) = 2^d$, and the VC dimension is $V_{\mathcal{A}} \geq d$.

Now we need to show that we cannot shatter $d + 1$ points. Let $z_1, \dots, z_{d+1} \in \mathbb{R}^d$. For any $i \in \{1, \dots, d\}$, we define $\bar{z}_i$ as a point among $\{z_1, \dots, z_{d+1}\}$ having the largest i -th coordinate, i.e. $\bar{z}_{i,i} = \max\{z_{1,i}, \dots, z_{d+1,i}\}$. Then, for any $x \in \mathbb{R}^d$, we have :

$$\begin{aligned} \{\bar{z}_1, \dots, \bar{z}_d\} \subset A_x &\iff \forall i, k \in \{1, \dots, d\}, \bar{z}_{i,k} \leq x_k \\ &\iff \forall k \in \{1, \dots, d\}, \bar{z}_{k,k} \leq x_k \quad (\bar{z}_{i,k} \leq \bar{z}_{k,k} \text{ by definition}) \\ &\iff \forall k \in \{1, \dots, d\}, z_{1,k}, \dots, z_{d+1,k} \leq x_k \\ &\iff \{z_1, \dots, z_{d+1}\} \subset A_x. \end{aligned}$$

Since $\{\bar{z}_1, \dots, \bar{z}_d\}$ contains at most d points, at least one of the z_j is not in it. Thus, we cannot find $A \in \mathcal{A}$ that isolates $\bar{z}_1, \dots, \bar{z}_d$ from the other points, and $S_{\mathcal{A}}(d+1) < 2^{d+1}$. By definition, the VC dimension is $V_{\mathcal{A}} = d$.

2)

For any $x = (x_1, \dots, x_d)$, $y = (y_1, \dots, y_d) \in \mathbb{R}^d$, we write $x \wedge y := (\min(x_1, y_1), \dots, \min(x_d, y_d))$ and $x \vee y := (\max(x_1, y_1), \dots, \max(x_d, y_d))$ for the coordinate-wise minimum and maximum. When $x_i \leq y_i$ for all i , we write $[x, y] := [x_1, y_1] \times \dots \times [x_d, y_d]$, so that $\mathcal{A} = \{[x, y] : x, y \in \mathbb{R}^d, x \wedge y = x\}$ (closed rectangles with sides parallel to the axes). We also write $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^d$. Let us show that $V_{\mathcal{A}} = 2d$.

$\mathcal{A}$ shatters $2d$ points. Consider the canonical basis vectors $e_1, \dots, e_d$ of $\mathbb{R}^d$ and their opposites $-e_1, \dots, -e_d$, and let $E := \{-e_1, \dots, -e_d, e_1, \dots, e_d\}$. Fix $\epsilon \in]0, 1[$.

(i) The empty subset : $E \cap [-\epsilon \mathbf{1}, \epsilon \mathbf{1}] = \emptyset$, since every point of E has one coordinate equal to ± 1 , and $|\pm 1| > \epsilon$.

(ii) A non-empty subset $S = \{s_1, \dots, s_m\} \subset E$: let

$$R_S := \left[\bigwedge_{k=1}^m s_k - \epsilon \mathbf{1}, \bigvee_{k=1}^m s_k + \epsilon \mathbf{1} \right] \in \mathcal{A}.$$

By construction, $S \subset R_S$. Now let $p \in E \setminus S$:

- if $p = e_j$, then $p_j = 1$. Since $e_j \notin S$, the j -th coordinate of every s_k is in $\{-1, 0\}$, so the upper bound of R_S in direction j is at most $\epsilon < 1$, and $p \notin R_S$,
- if $p = -e_j$, then $p_j = -1$. Since $-e_j \notin S$, the j -th coordinate of every s_k is in $\{0, 1\}$, so the lower bound of R_S in direction j is at least $-\epsilon > -1$, and $p \notin R_S$.

Thus, $E \cap R_S = S$.

Every subset of E can be picked out by a rectangle, so $N_{\mathcal{A}}(-e_1, \dots, -e_d, e_1, \dots, e_d) = 2^{2d}$, $S_{\mathcal{A}}(2d) = 2^{2d}$, and $V_{\mathcal{A}} \geq 2d$.

$\mathcal{A}$ does not shatter $2d + 1$ points. Let $z_1, \dots, z_{2d+1} \in \mathbb{R}^d$, with $z_j = (z_{j,1}, \dots, z_{j,d})$. For any $i \in \{1, \dots, d\}$, we define $\bar{z}_i$ (resp. $\underline{z}_i$) as a point among $\{z_1, \dots, z_{2d+1}\}$ having the largest (resp. smallest) i -th coordinate, i.e. $\bar{z}_{i,i} = \max\{z_{1,i}, \dots, z_{2d+1,i}\}$ and $\underline{z}_{i,i} = \min\{z_{1,i}, \dots, z_{2d+1,i}\}$. Let $T := \{\underline{z}_1, \dots, \underline{z}_d, \bar{z}_1, \dots, \bar{z}_d\}$. Then, for any rectangle $[x, y] \in \mathcal{A}$:

$$\begin{aligned} T \subset [x, y] &\iff \forall i, k \in \{1, \dots, d\}, \underline{z}_{i,k}, \bar{z}_{i,k} \in [x_k, y_k] \\ &\iff \forall k \in \{1, \dots, d\}, \underline{z}_{k,k}, \bar{z}_{k,k} \in [x_k, y_k] \quad (\underline{z}_{k,k} \leq \underline{z}_{i,k}, \bar{z}_{i,k} \leq \bar{z}_{k,k}) \\ &\iff \forall k \in \{1, \dots, d\}, z_{1,k}, \dots, z_{2d+1,k} \in [x_k, y_k] \\ &\iff \{z_1, \dots, z_{2d+1}\} \subset [x, y]. \end{aligned}$$

Since T contains at most $2d$ points, at least one of the z_j is not in it. In other words, no rectangle picks out exactly the subset T , so $S_{\mathcal{A}}(2d + 1) < 2^{2d+1}$.

By definition of the VC dimension, we conclude that $V_{\mathcal{A}} = 2d$.

EXERCISE 4. Give an upper bound for the VC dimension of the class of closed balls in $\mathbb{R}^d$:

$$\mathcal{A} = \left\{ \left\{ x = (x_1, \dots, x_d) \in \mathbb{R}^d : \sum_{i=1}^d |x_i - a_i|^2 \leq b \right\} : a_1, \dots, a_d, b \in \mathbb{R} \right\}.$$

Correction Exercise 4

To answer the question, we first prove the following theorem.

Let $\mathcal{F}$ be a vector space of functions $\mathbb{R}^d \rightarrow \mathbb{R}$ of finite dimension $m \in \mathbb{N}^$. Then the class of sets $\mathcal{B} := \{ \{x \in \mathbb{R}^d : f(x) \geq 0\} : f \in \mathcal{F} \}$ has VC dimension $V_{\mathcal{B}} \leq m$.*

It suffices to show that no set of $m + 1$ points of $\mathbb{R}^d$ can be shattered by $\mathcal{B}$. Let $x_1, \dots, x_{m+1} \in \mathbb{R}^d$, and consider the linear map

$$h : f \in \mathcal{F} \mapsto (f(x_1), \dots, f(x_{m+1})) \in \mathbb{R}^{m+1}.$$

Since $\mathcal{F}$ is a vector space and h is linear, $h(\mathcal{F})$ is a vector subspace of $\mathbb{R}^{m+1}$ of dimension at most $\dim \mathcal{F} = m < m + 1$. Thus, its orthogonal complement is not reduced to $\{0\}$: there exists a non-zero vector $\gamma = (\gamma_1, \dots, \gamma_{m+1}) \in \mathbb{R}^{m+1}$ orthogonal to $h(\mathcal{F})$, i.e. such that for every $f \in \mathcal{F}$:

$$\langle \gamma, h(f) \rangle = \gamma_1 f(x_1) + \dots + \gamma_{m+1} f(x_{m+1}) = 0. \quad (6)$$

Let $P := \{i \in \llbracket 1, m+1 \rrbracket : \gamma_i \geq 0\}$ and $N := \{i \in \llbracket 1, m+1 \rrbracket : \gamma_i < 0\}$. Then, for every $f \in \mathcal{F}$:

$$\sum_{i \in P} \gamma_i f(x_i) = - \sum_{i \in N} \gamma_i f(x_i). \quad (7)$$

Up to replacing γ by $-\gamma$ (which also satisfies (6)), we can assume $N \neq \emptyset$ (since $\gamma \neq 0$, either γ or $-\gamma$ has a negative coordinate).

Now suppose, by contradiction, that there exists $f \in \mathcal{F}$ such that $\{x \in \mathbb{R}^d : f(x) \geq 0\}$ contains $\{x_i : i \in P\}$ but none of the points $\{x_i : i \in N\}$. Then :

- for $i \in P$, $\gamma_i \geq 0$ and $f(x_i) \geq 0$, so the left-hand side of (7) is non-negative,
- for $i \in N$, $\gamma_i < 0$ and $f(x_i) < 0$, so $\gamma_i f(x_i) > 0$, and since $N \neq \emptyset$ the right-hand side of (7) is negative,

which is a contradiction. Thus, no set of $\mathcal{B}$ picks out exactly $\{x_i : i \in P\}$ among $\{x_1, \dots, x_{m+1}\}$, so $\mathcal{B}$ does not shatter these points. Since the points were arbitrary, $S_{\mathcal{B}}(m+1) < 2^{m+1}$, and $V_{\mathcal{B}} \leq m$.

Let us now go back to the question. For any $a_1, \dots, a_d, b \in \mathbb{R}$ and $x = (x_1, \dots, x_d) \in \mathbb{R}^d$, we have :

$$\sum_{i=1}^d |x_i - a_i|^2 - b = \sum_{i=1}^d x_i^2 - 2 \sum_{i=1}^d a_i x_i + \left(\sum_{i=1}^d a_i^2 - b \right).$$

Let $\mathcal{F}$ be the vector space spanned by the functions $\phi_1, \dots, \phi_{d+2} : \mathbb{R}^d \rightarrow \mathbb{R}$ defined for every $x = (x_1, \dots, x_d) \in \mathbb{R}^d$ by

$$\phi_1(x) = \sum_{i=1}^d x_i^2, \quad \phi_2(x) = x_1, \quad \dots, \quad \phi_{d+1}(x) = x_d, \quad \phi_{d+2}(x) = 1.$$

Then $\dim \mathcal{F} \leq d+2$, and every ball of $\mathcal{A}$ can be written $\{x \in \mathbb{R}^d : f(x) \leq 0\}$ with $f = \phi_1 - 2 \sum_{i=1}^d a_i \phi_{i+1} + (\sum_{i=1}^d a_i^2 - b) \phi_{d+2} \in \mathcal{F}$. Hence :

$$\mathcal{A} \subset \{ \{x \in \mathbb{R}^d : f(x) \leq 0\} : f \in \mathcal{F} \} = \{ \{x \in \mathbb{R}^d : -f(x) \geq 0\} : f \in \mathcal{F} \} = \mathcal{B},$$

where the last equality holds because $f \in \mathcal{F} \iff -f \in \mathcal{F}$ ($\mathcal{F}$ is a vector space). Since a subclass cannot shatter more points than the whole class, applying the theorem gives :

$$V_{\mathcal{A}} \leq V_{\mathcal{B}} \leq d+2.$$

EXERCISE 5. Let $\mathcal{A}$ be a class of sets in $\mathbb{R}^d$ with finite VC dimension $V < +\infty$ and shattering coefficients $s(\mathcal{A}, n)$, for every $n \in \mathbb{N}^*$.

- 1) Show that : $\forall n \geq 1, s(\mathcal{A}, n) \leq (n+1)^V$.
- 2) Show that : $\forall n \geq V, s(\mathcal{A}, n) \leq (ne/V)^V$.

Hint. Use Sauer's lemma : $\forall n \geq 1, s(\mathcal{A}, n) \leq \sum_{k=0}^V \binom{n}{k}$.

Correction Exercise 5

1) Let $n \in \mathbb{N}^*$. By the binomial formula :

$$(n+1)^V = \sum_{k=0}^V \binom{V}{k} n^k = \sum_{k=0}^V \frac{V(V-1) \cdots (V-k+1)}{k!} n^k.$$

For every $k \in \llbracket 0, V \rrbracket$, we have :

- $V(V-1) \cdots (V-k+1) \geq 1$ (product of k positive integers, equal to 1 if $k = 0$),
- $n^k \geq n(n-1) \cdots (n-k+1)$ (each factor on the right is at most n ; if $k > n$, the right-hand side is 0).

Thus :

$$(n+1)^V \geq \sum_{k=0}^V \frac{n(n-1) \cdots (n-k+1)}{k!} = \sum_{k=0}^V \binom{n}{k} \geq s(\mathcal{A}, n) \quad (\text{Sauer's lemma}).$$

2) We assume $V \geq 1$ (otherwise $(ne/V)^V$ is not defined). Let $n \geq V$ be an integer, so that $V/n \leq 1$. By Sauer's lemma, it is enough to show that

$$\sum_{k=0}^V \binom{n}{k} \leq \left(\frac{ne}{V}\right)^V = \left(\frac{n}{V}\right)^V e^V, \quad \text{i.e.} \quad \left(\frac{V}{n}\right)^V \sum_{k=0}^V \binom{n}{k} \leq e^V.$$

We have :

$$\begin{aligned} \left(\frac{V}{n}\right)^V \sum_{k=0}^V \binom{n}{k} &\leq \sum_{k=0}^V \binom{n}{k} \left(\frac{V}{n}\right)^k && (V/n \leq 1 \text{ and } k \leq V, \text{ so } (V/n)^V \leq (V/n)^k) \\ &\leq \sum_{k=0}^n \binom{n}{k} \left(\frac{V}{n}\right)^k && (n \geq V \text{ and all terms are non-negative}) \\ &= \left(1 + \frac{V}{n}\right)^n && (\text{binomial formula}) \\ &\leq \left(e^{V/n}\right)^n = e^V && (1+x \leq e^x, \text{ by convexity of the exponential}). \end{aligned}$$

Hence $s(\mathcal{A}, n) \leq \sum_{k=0}^V \binom{n}{k} \leq (ne/V)^V$.